Tasks

1. **Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.**

   ➢

   1. **Download & Install FileZilla Server**
      - Go to the official FileZilla website and download *FileZilla Server* for Windows.
      - Run the installer and follow the setup wizard.
      - After installation, open *FileZilla Server Interface*.
   2. **Create a User Account**
      - In the FileZilla Server interface, go to Edit → Users.
      - Click Add and enter a username (e.g., *student1*).
      - Set a password for the user under the General tab.
   3. **Set Up Shared Folder**
      - In the same user settings window, go to Shared folders.
      - Click Add and select a folder from your computer (e.g., *D:\FTPShare*).
      - Assign permissions like *Read*, *Write*, *Delete*, and *Append* depending on requirements.
   4. **Test FTP Access**
      - Open any FTP client (like FileZilla Client) or use the browser.
      - Connect using your server's IP address, the username, and password.
      - Verify that the shared folder contents are accessible.

2. **Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.**

   ➢

   1. **Open FileZilla Client**
      - Launch *FileZilla Client* on your Windows machine.
      - In the top bar, enter the following details:
        - Host: Your server's IP address (e.g., 127.0.0.1 for local).
        - Username: The FTP user account you created (e.g., *student1*).
        - Password: The password set for that user.
        - Port: 21 (default FTP port).
   2. **Connect to the Server**
      - Click Quickconnect.
      - The status window will show connection messages.
      - Once connected, the left panel shows your local files, and the right panel shows the server's shared folder.
   3. **Upload an Image File**
      - From the left panel, browse to any image file on your computer (e.g., *C:\Users\Pictures\sample.jpg*).
      - Drag and drop the file into the right panel (server shared folder).
      - The transfer queue at the bottom will confirm successful upload.
   4. **Verify Upload**
      - Check the right panel to ensure the image file appears in the shared folder.
      - Optionally, reconnect or refresh to confirm the file is accessible from the server.

3. **Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.**

➢

1. **Connect to FTP Server**
   - Open *FileZilla Client* on your Windows machine.
   - Enter the server details in the top bar:
     - Host: Your server's IP address
     - Username: The FTP account you created
     - Password: The set password.
     - Port: 21.
   - **Click Quickconnect to establish the connection.**
2. **Locate Uploaded File**
   - In the right panel (server side), navigate to the shared folder.
   - Identify the image file you previously uploaded
3. **Download File to Different Folder**
   - In the left panel browse to a different folder on your computer
   - Drag the file from the right panel to the left panel
   - The transfer queue at the bottom will confirm successful download.
4. **Verify File Size**
   - On your computer, right-click the downloaded file and select Properties.
   - Compare the file size with the original image file.
   - Ensure both sizes match, confirming the file integrity.

4. **Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).

Hint: Think about how network admins monitor device health, bandwidth, or uptime.**

➢

Scenario	How SNMP is Used	Example in Daily Life
1. ISP Backbone Monitoring	ISPs use SNMP to track router/switch interface status, bandwidth utilization, and error rates. SNMP traps alert admins if a link fails or traffic exceeds thresholds.	When your broadband provider ensures stable internet by monitoring backbone routers and quickly fixing outages.
2. Campus WiFi & LAN Management	University IT teams use SNMP to monitor access points, switches, and firewalls. It helps detect overloaded APs, failed ports, or high CPU usage.	When students complain of slow WiFi, admins check SNMP dashboards to see if an AP is overloaded or a switch port is down.
3. Large Website & Data Center Operations	SNMP monitors servers, UPS systems, and load balancers. It provides real-time alerts about hardware health, bandwidth spikes, or device failures.	Websites like e-commerce platforms use SNMP to ensure uptime—if a load balancer fails, SNMP alerts trigger automated failover.

5. **Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.**

➢

Aspect	FTP (File Transfer Protocol)	HTTP (Hypertext Transfer Protocol)
Port Number	Default port 21 (control), port 20 (data)	Default port 80 (HTTP), 443 (HTTPS)
Security	Plain text by default, insecure; can use FTPS/SFTP for encryption	HTTP is insecure by default; HTTPS adds SSL/TLS encryption
Data Transfer Method	Separate control and data channels; supports both active and passive modes	Single request–response model over one connection
Use Cases	Uploading/downloading files between client and server (e.g., website file hosting, backups)	Browsing websites, accessing web apps, APIs, streaming content
Authentication	Requires username & password (can be anonymous FTP)	Usually no authentication for public sites; can use cookies, tokens, or basic auth for restricted access